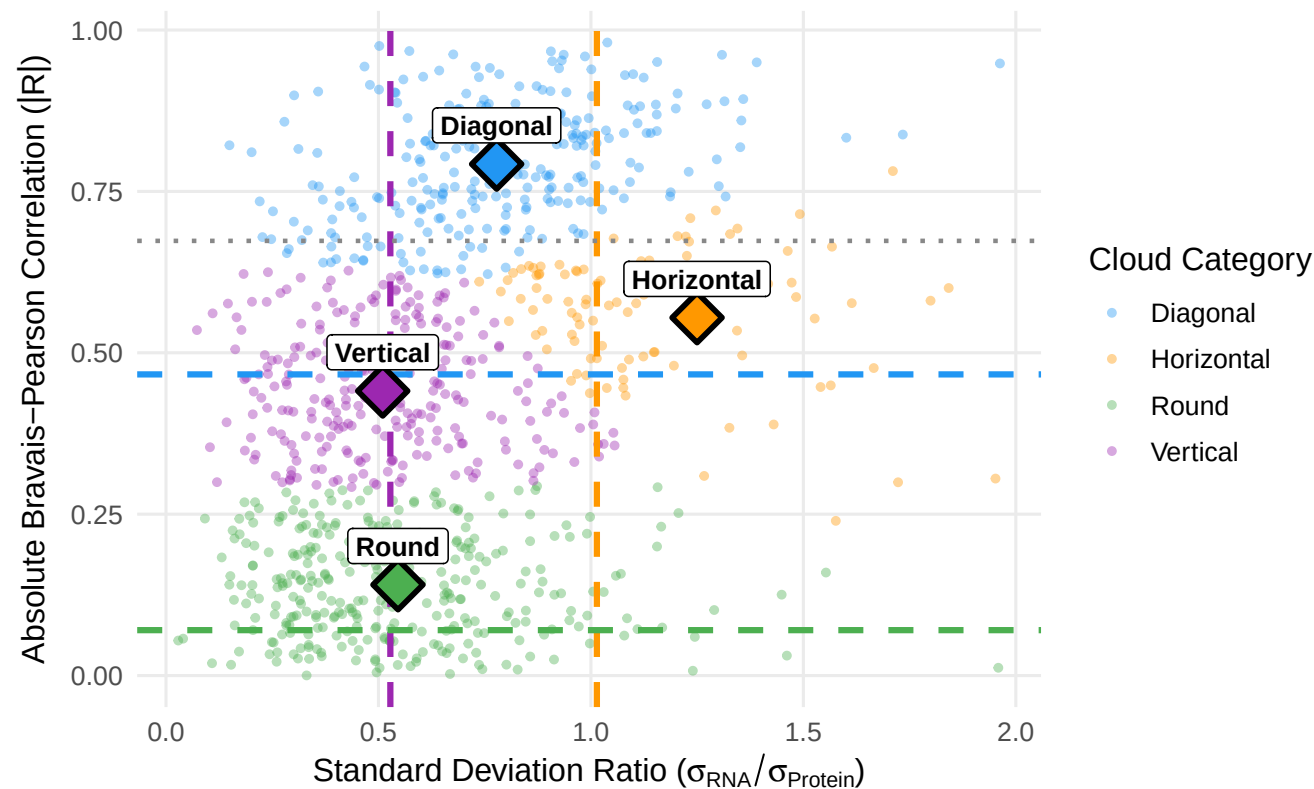


Data-Driven Multi-Omics Cloud Morphology Classification (2D K-Means, BDL Cohort)

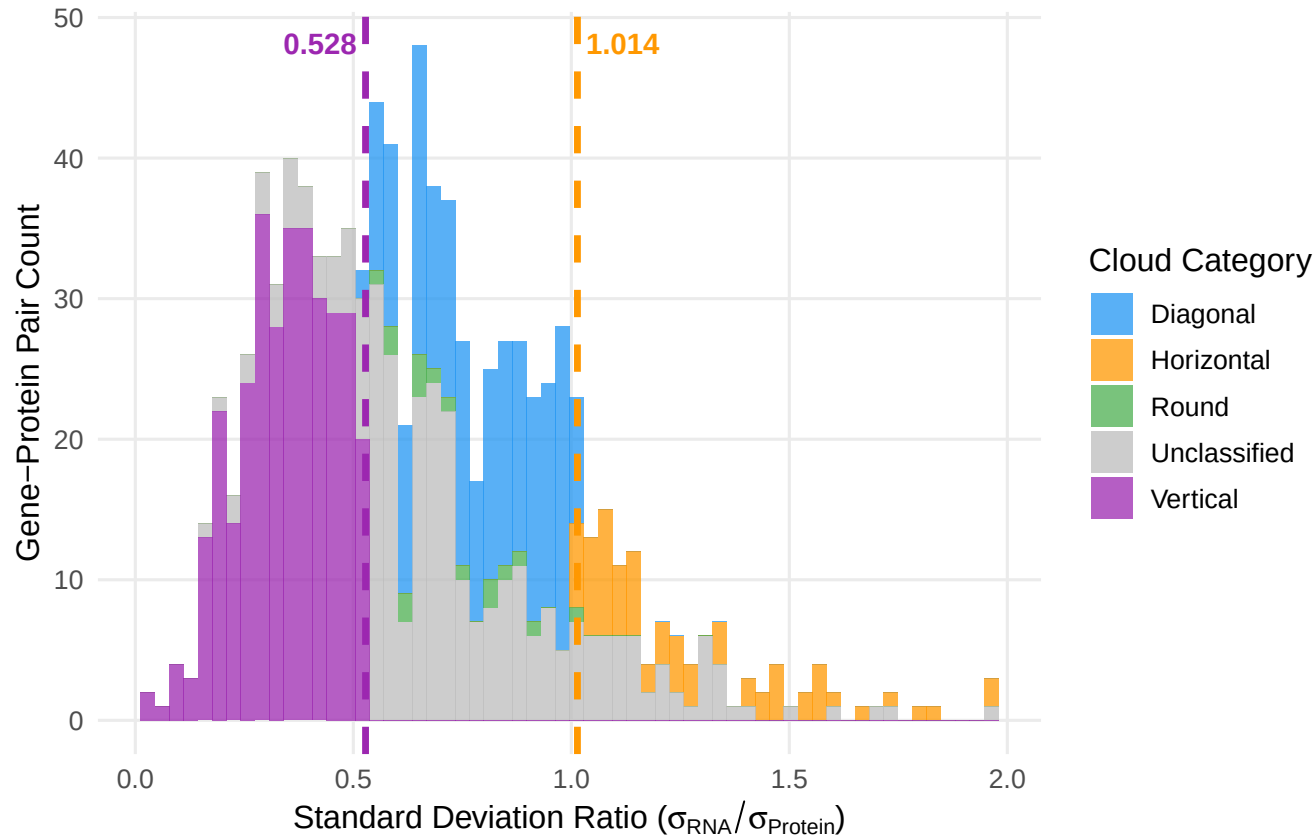
Decision boundaries represent objective midpoints between adjacent unsupervised cluster centroids

A. Unsupervised 2D K-Means Partitioning in (Ratio, |R|) Space

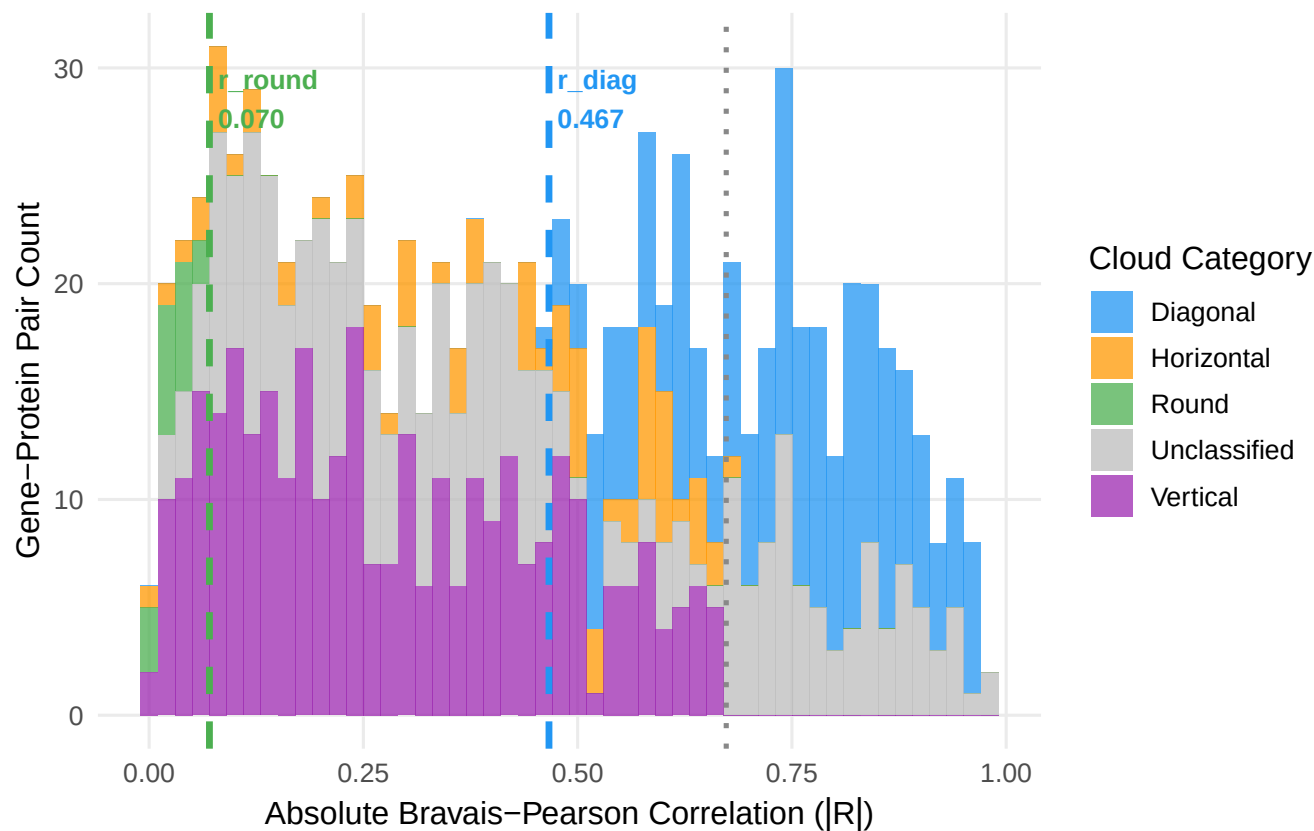
Diamonds indicate cluster centroids. Thresholds: ratio_low=0.528, ratio_high=1.014, r



B. Distribution of Variance Ratio with Decision Boundaries



C. Distribution of Correlation Magnitudes with Thresholds



D. Empirical Multi-Omics Category Frequencies

